

The Accuracy of the Magnus Expansion for Polarization-Mode Dispersion and Polarization-Dependent Loss

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Abstract—We employ the Magnus expansion to solve the differential equation for the frequency dependence of the Mueller matrix in the presence of polarization-mode dispersion and polarization-dependent loss. We then compare our solution with the results of previous authors and establish its accuracy to fifth order in frequency.

Index Terms—Communication systems, optical fibers, polarization-mode dispersion (PMD).

I. INTRODUCTION

THE input and output Stokes vectors [1] in a linear optical system such as a fiber or waveguide are related by a rotation matrix in the absence of polarization-dependent loss (PDL). This matrix is often described by an expansion in polarization-mode dispersion (PMD) vectors of increasing order [2]. In this paper, we generalize this procedure to the 4×4 Mueller matrix transformation for systems with both PMD and PDL. This matrix obeys a simple differential equation involving only the real and imaginary components of the complex principal state vector [3]–[6] that we solve with the Magnus expansion. Unlike other procedures, truncating this solution yields approximations that preserve the group properties of the exact expression [7]–[10]. In the absence of PDL, we further demonstrate that the Magnus expansion [3] agrees with previous power series formulations [1], [2], [11] for the frequency variation of the Stokes vector, while a detailed numerical analysis indicates that for a specified model order the Magnus expansion maximizes the bandwidth of high Jones matrix estimation accuracy and, therefore, can efficiently invert the transformation between the frequency-dependent PMD vector and the Jones/Mueller matrix. We also discuss the relationship between higher-order PMD and PDL effects, which we describe with both 4×4 Mueller and 2×2 Jones matrices.

II. MUELLER AND JONES MATRIX FORMALISMS

A. Magnus Expansion

In a linear optical system characterized by two orthogonally polarized guided modes with different group velocities and

losses (PMD and PDL), the Jones vectors of the output and the frequency-independent input electric field polarizations, $|t(\omega)\rangle$ and $|s\rangle$, at an optical frequency ω are related through a 2×2 complex transfer matrix $|t(\omega)\rangle = \mathbf{T}(\omega)|s\rangle$ (as this paper only considers attenuation and phase differences between the two polarizations, we further set $\det(\mathbf{T}) = 1$ [5]). The matrix $\mathbf{T}(\omega)$ satisfies [12]

$$\frac{d\mathbf{T}}{d\omega}\mathbf{T}^{-1} = -\frac{i}{2}\vec{W}(\omega) \cdot \vec{\sigma} \quad (1)$$

in which $\vec{W} = \vec{\Omega} + i\vec{\Lambda}$ is termed the complex principal state vector. The corresponding input and output 4×1 Stokes vectors for a completely polarized incident electric field $\tilde{s} = [\langle s|s\rangle, \langle s|\vec{\sigma}|s\rangle]^T$ and $\tilde{t}(\omega) = [\langle t|t\rangle, \langle t|\vec{\sigma}|t\rangle]^T$ are similarly related through $\tilde{t}(\omega) = \mathbf{M}(\omega)\tilde{s}$ in which $\mathbf{M}_{nm} = \text{Trace}(\sigma_n \mathbf{T} \sigma_m \mathbf{T}^\dagger)/2$ designate the components of the 4×4 Mueller matrix $\mathbf{M}(\omega)$, $n, m = 0, 1, 2, 3$, σ_0 denotes 2×2 identity matrix and $\vec{\sigma} = [\sigma_1, \sigma_2, \sigma_3]^T$ is the vector of Pauli spin matrices [1], [13].

Alternatively, the Mueller matrix can be expressed as a Kronecker matrix product [3], [14]–[16]

$$\mathbf{M}(\omega) = \mathbf{A} [\mathbf{T}(\omega) \otimes \mathbf{T}^*(\omega)] \mathbf{A}^\dagger \quad (2)$$

in which

$$\mathbf{A} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & -1 \\ 0 & 1 & 1 & 0 \\ 0 & i & -i & 0 \end{bmatrix} \quad (3)$$

is unitary, while $*$ and $\dagger$ represent complex and hermitian conjugation, respectively. If we define for any two vectors $\vec{a}$ and $\vec{b}$ [17]

$$\begin{bmatrix} 0 & \vec{a}^T \\ \vec{a} & \vec{b} \times \end{bmatrix} = \begin{bmatrix} 0 & a_x & a_y & a_z \\ a_x & 0 & -b_z & b_y \\ a_y & b_z & 0 & -b_x \\ a_z & -b_y & b_x & 0 \end{bmatrix} \quad (4)$$

we then have

$$\mathbf{A}[(\vec{a} \cdot \vec{\sigma}) \otimes \sigma_0] \mathbf{A}^\dagger = \begin{bmatrix} 0 & \vec{a}^T \\ \vec{a} & i\vec{a} \times \end{bmatrix} \quad (5)$$

while

$$\mathbf{A}[\sigma_0 \otimes (\vec{a} \cdot \vec{\sigma})^*] \mathbf{A}^\dagger = \begin{bmatrix} 0 & \vec{a}^T \\ \vec{a} & -i\vec{a} \times \end{bmatrix}. \quad (6)$$

Manuscript received March 6, 2008; revised June 20, 2008. Current version published December 19, 2008. This work was supported in part by the National Sciences and Research Council of Canada, in part by Nortel Networks, in part by the Center for Electrophotonic Materials and Devices (CEMD), and in part by the Ontario Research and Development Challenge Fund (ORDCF).

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Digital Object Identifier 10.1109/JLT.2008.928928

Accordingly, in terms of $\vec{\Omega}$ and $\vec{\Lambda}$ differentiation of (2) yields [3], [4], [6]

$$\frac{d\mathbf{M}}{d\omega}\mathbf{M}^{-1} = \mathbf{H}(\omega) \quad (7)$$

$$\mathbf{H}(\omega) = \begin{bmatrix} 0 & \vec{\Lambda}^T \\ \vec{\Lambda} & \vec{\Omega} \times \end{bmatrix}. \quad (8)$$

The matrix $\mathbf{H}(\omega)$ can then be considered a generator of an infinitesimal Lorentz transformation in Stokes space [15], [17], [18].

The variation of $\mathbf{M}(\omega)$ with frequency can now be determined iteratively in terms of $\mathbf{M}(\omega_0)$ at the optical carrier frequency, ω_0 , according to [9], [19]

$$\mathbf{M}(\omega) = \mathbf{M}(\omega_0) + \int_{\omega_0}^{\omega} d\omega_1 \mathbf{H}(\omega_1) \mathbf{M}(\omega_1). \quad (9)$$

This yields

$$\mathbf{M}(\omega) = [\mathbf{I}_N + \mathbf{J}_1 + \mathbf{J}_2 + \mathbf{J}_3 + \dots] \mathbf{M}(\omega_0) \quad (10)$$

where $\mathbf{I}_N$ is the $N \times N$ identity matrix and

$$\mathbf{J}_n(\omega) = \int_{\omega_0}^{\omega} d\omega_1 \dots \int_{\omega_0}^{\omega_{n-1}} d\omega_n \mathbf{H}(\omega_1) \dots \mathbf{H}(\omega_n) \quad (11)$$

are the series solution coefficients of (7). On the other hand, as demonstrated in subsequent sections, the frequency interval over which (10) approximates $\mathbf{M}$ to within a given level of accuracy is improved if we instead truncate the Magnus expansion

$$\mathbf{M}(\omega) = \exp[\mathbf{B}_1 + \mathbf{B}_2 + \mathbf{B}_3 + \dots] \mathbf{M}(\omega_0) \quad (12)$$

to a given number of terms, which preserves relevant group properties of the exact solution [3], [7]–[9]. To fourth order in $\mathbf{H}$ the expansion coefficients $\mathbf{J}_n$ and $\mathbf{B}_n$ of (10) and (12) are related by

$$\begin{aligned} \mathbf{B}_1 &= \mathbf{J}_1 \\ \mathbf{B}_2 &= \mathbf{J}_2 - \frac{\mathbf{J}_1^2}{2} \\ \mathbf{B}_3 &= \mathbf{J}_3 - \frac{1}{2}(\mathbf{J}_1\mathbf{J}_2 + \mathbf{J}_2\mathbf{J}_1) + \frac{\mathbf{J}_1^3}{3} \\ \mathbf{B}_4 &= \mathbf{J}_4 - \frac{1}{2}(\mathbf{J}_1\mathbf{J}_3 + \mathbf{J}_3\mathbf{J}_1 + \mathbf{J}_2\mathbf{J}_2) \\ &\quad + \frac{1}{3}(\mathbf{J}_1\mathbf{J}_1\mathbf{J}_2 + \mathbf{J}_1\mathbf{J}_2\mathbf{J}_1 + \mathbf{J}_2\mathbf{J}_1\mathbf{J}_1) - \frac{\mathbf{J}_1^4}{4} \end{aligned} \quad (13)$$

with analogous formulas for higher orders [10]. Rewriting $\mathbf{B}_n$ in terms of the commutators $[\mathbf{H}(\omega_i), \mathbf{H}(\omega_j)]$ of $\mathbf{H}(\omega)$ at two optical frequencies ω_i and ω_j yields an alternative, computationally convenient formula for the Magnus expansion coefficients [8], [10]

$$\begin{aligned} \mathbf{B}_1 &= \int_{\omega_0}^{\omega} d\omega_1 \mathbf{H}(\omega_1) \\ \mathbf{B}_2 &= \frac{1}{2} \int_{\omega_0}^{\omega} d\omega_1 \int_{\omega_0}^{\omega_1} d\omega_2 [\mathbf{H}(\omega_1), \mathbf{H}(\omega_2)] \\ \mathbf{B}_3 &= \frac{1}{6} \int_{\omega_0}^{\omega} d\omega_1 \int_{\omega_0}^{\omega_1} d\omega_2 \int_{\omega_0}^{\omega_2} d\omega_3 \\ &\quad \times \left([\mathbf{H}(\omega_1), [\mathbf{H}(\omega_2), \mathbf{H}(\omega_3)]] \right. \\ &\quad \left. + [[\mathbf{H}(\omega_1), \mathbf{H}(\omega_2)], \mathbf{H}(\omega_3)] \right) \end{aligned}$$

$$\begin{aligned} \mathbf{B}_4 &= \frac{1}{12} \int_{\omega_0}^{\omega} d\omega_1 \int_{\omega_0}^{\omega_1} d\omega_2 \int_{\omega_0}^{\omega_2} d\omega_3 \int_{\omega_0}^{\omega_3} d\omega_4 \\ &\quad \times \left([\mathbf{H}(\omega_1), [[\mathbf{H}(\omega_2), \mathbf{H}(\omega_3)], \mathbf{H}(\omega_4)]] \right. \\ &\quad \left. + [[\mathbf{H}(\omega_1), [\mathbf{H}(\omega_2), \mathbf{H}(\omega_3)]], \mathbf{H}(\omega_4)] \right. \\ &\quad \left. + [[\mathbf{H}(\omega_1), \mathbf{H}(\omega_2)], [\mathbf{H}(\omega_3), \mathbf{H}(\omega_4)]] \right). \end{aligned} \quad (14)$$

If the matrix $\mathbf{H}(\omega)$ is now expanded in a Taylor series

$$\mathbf{H}(\omega) = \sum_{n=0}^{\infty} \mathbf{H}_n \frac{\Delta\omega^n}{n!} \quad (15)$$

where the coefficient $\mathbf{H}_n = [d^n \mathbf{H} / d\omega^n]_{\omega_0}$ is the n th order PMD/PDL matrix, we obtain to fifth order in $\Delta\omega = \omega - \omega_0$ from the Magnus coefficients, (14),

$$\begin{aligned} \mathbf{M}(\omega)\mathbf{M}(\omega_0)^{-1} &= \exp[\mathbf{N}(\Delta\omega)] \\ \mathbf{N}(\Delta\omega) &= \mathbf{H}_0\Delta\omega + \mathbf{H}_1 \frac{\Delta\omega^2}{2!} \\ &\quad + \left(\mathbf{H}_2 - \frac{1}{2}[\mathbf{H}_0, \mathbf{H}_1] \right) \frac{\Delta\omega^3}{3!} \\ &\quad + \left(\mathbf{H}_3 - [\mathbf{H}_0, \mathbf{H}_2] \right) \frac{\Delta\omega^4}{4!} \\ &\quad + \left(\mathbf{H}_4 - [\mathbf{H}_1, \mathbf{H}_2] - \frac{3}{2}[\mathbf{H}_0, \mathbf{H}_3] \right. \\ &\quad \left. + \frac{1}{6}[\mathbf{H}_0, [\mathbf{H}_0, \mathbf{H}_2]] \right. \\ &\quad \left. + \frac{1}{2}[\mathbf{H}_1, [\mathbf{H}_1, \mathbf{H}_0]] \right. \\ &\quad \left. + \frac{1}{6}[\mathbf{H}_0, [\mathbf{H}_0, [\mathbf{H}_0, \mathbf{H}_1]]] \right) \frac{\Delta\omega^5}{5!} \\ &\quad + O(\Delta\omega^6). \end{aligned} \quad (16)$$

Alternatively, (8) expresses the Mueller matrix directly in terms of the vectors $\vec{\Omega}$ and $\vec{\Lambda}$. To fourth order

$$\mathbf{M}(\omega) = \exp \begin{bmatrix} 0 & \vec{d}^T \\ \vec{a} & \vec{b} \times \end{bmatrix} \mathbf{M}(\omega_0). \quad (17)$$

in which $\vec{a}$ and $\vec{b}$ designate

$$\begin{aligned} \vec{a} &= \vec{\Lambda}_0\Delta\omega + \vec{\Lambda}_1 \frac{\Delta\omega^2}{2!} \\ &\quad + \left(\vec{\Lambda}_2 - \frac{1}{2}\vec{\Omega}_0 \times \vec{\Lambda}_1 - \frac{1}{2}\vec{\Lambda}_0 \times \vec{\Omega}_1 \right) \frac{\Delta\omega^3}{3!} \\ &\quad + \left(\vec{\Lambda}_3 - \vec{\Omega}_0 \times \vec{\Lambda}_2 - \vec{\Lambda}_0 \times \vec{\Omega}_2 \right) \frac{\Delta\omega^4}{4!} + O(\Delta\omega^5) \end{aligned} \quad (18)$$

and

$$\begin{aligned} \vec{b} &= \vec{\Omega}_0\Delta\omega + \vec{\Omega}_1 \frac{\Delta\omega^2}{2!} \\ &\quad + \left(\vec{\Omega}_2 - \frac{1}{2}\vec{\Omega}_0 \times \vec{\Omega}_1 - \frac{1}{2}\vec{\Lambda}_1 \times \vec{\Lambda}_0 \right) \frac{\Delta\omega^3}{3!} \\ &\quad + \left(\vec{\Omega}_3 - \vec{\Omega}_0 \times \vec{\Omega}_2 - \vec{\Lambda}_2 \times \vec{\Lambda}_0 \right) \frac{\Delta\omega^4}{4!} + O(\Delta\omega^5) \end{aligned} \quad (19)$$

where the matrix exponential of (17) can be computed efficiently by employing the formalism of [3]. Note that $\mathbf{M}(\omega)$ as given by (17)–(19) includes previously unnoticed correlations between the Taylor orders of $\vec{\Lambda}$ and $\vec{\Omega}$.

Additionally, since any Jones matrix of the form $\mathbf{T}(\omega) = \exp[\mathbf{K}]\mathbf{T}(\omega_0)$, for a given 2×2 complex matrix $\mathbf{K}$, can be transformed according to (2) into the Mueller matrix $\mathbf{M}(\omega) = \exp[\mathbf{A}(\mathbf{K} \otimes \boldsymbol{\sigma}_0 + \boldsymbol{\sigma}_0 \otimes \mathbf{K}^*)\mathbf{A}^\dagger]\mathbf{M}(\omega_0)$ [3], [15], we conclude from (16) that

$$\begin{aligned} \mathbf{T}(\omega)\mathbf{T}(\omega_0)^{-1} &= \exp\left[-\frac{i}{2}\vec{n}(\Delta\omega) \cdot \vec{\sigma}\right] \\ \vec{n}(\Delta\omega) &= \vec{W}_0\Delta\omega + \vec{W}_1\frac{\Delta\omega^2}{2!} \\ &\quad + \left(\vec{W}_2 - \frac{1}{2}\vec{W}_0 \times \vec{W}_1\right)\frac{\Delta\omega^3}{3!} \\ &\quad + \left(\vec{W}_3 - \vec{W}_0 \times \vec{W}_2\right)\frac{\Delta\omega^4}{4!} \\ &\quad + \left(\vec{W}_4 - \vec{W}_1 \times \vec{W}_2 - \frac{3}{2}\vec{W}_0 \times \vec{W}_3 \right. \\ &\quad \left. + \frac{1}{6}\vec{W}_0 \times (\vec{W}_0 \times \vec{W}_2) \right. \\ &\quad \left. + \frac{1}{2}\vec{W}_1 \times (\vec{W}_1 \times \vec{W}_0) \right. \\ &\quad \left. + \frac{1}{6}\vec{W}_0 \times (\vec{W}_0 \times (\vec{W}_0 \times \vec{W}_1))\right)\frac{\Delta\omega^5}{5!} \\ &\quad + O(\Delta\omega^6) \end{aligned} \quad (20)$$

in which $\vec{W}_n = [d^n \vec{W}/d\omega^n]_{\omega_0}$, or, in terms of $\vec{\Omega}$ and $\vec{\Lambda}$, cf. (18), (19)

$$\mathbf{T}(\omega) = \exp\left[-\frac{i}{2}(\vec{b} + i\vec{a}) \cdot \vec{\sigma}\right]\mathbf{T}(\omega_0). \quad (21)$$

B. Exponential Series

In [11], the Jones matrix associated with PMD was evaluated to fourth-order in the frequency offset $\Delta\omega$. While, as we will show below, this procedure is equivalent to the expansion of (21), this series when truncated to finite order is nonunitary in the absence of PDL and, therefore, is accurate only within a somewhat reduced frequency interval.

Specializing to the case of zero PDL, and setting $\vec{b} = b\hat{b}$ with $\vec{\Lambda} = 0$, (21) can be written in terms of the unitary Jones matrix

$$\begin{aligned} \mathbf{U}(\Delta\omega) &\equiv \mathbf{T}(\omega)\mathbf{T}(\omega_0)^{-1} \\ &= \cos\left(\frac{b}{2}\right)\boldsymbol{\sigma}_0 - i\sin\left(\frac{b}{2}\right)(\hat{b} \cdot \vec{\sigma}). \end{aligned} \quad (22)$$

Next, we introduce the vector $\vec{u}$ and scalar u_0 , components of the Stokes vector formed from the elements $\mathbf{U}_{nm}$, $n, m = 0, 1$ of the Jones matrix $\mathbf{U}$ according to $u_0 = \text{Re}(\mathbf{U}_{00}) = \cos(b/2)$ and $\vec{u} = -[\text{Im}\{\mathbf{U}_{00}\}, \text{Im}\{\mathbf{U}_{01}\}, \text{Re}\{\mathbf{U}_{01}\}]^T = \sin(b/2)\hat{b}$.

In terms of the unit vector $\hat{p} \equiv \vec{\Omega}_0/|\vec{\Omega}_0|$, we find after expanding the quantities u_0 and $\vec{u}$ in a power series, cf. (19), and

identifying terms of equal orders in $\Delta\omega$

$$\begin{aligned} u_0 &= 1 - |\vec{\Omega}_0|^2 \frac{\Delta\omega^2}{8} - \vec{\Omega}_0 \cdot \vec{\Omega}_1 \frac{\Delta\omega^3}{8} \\ &\quad + \left(\frac{1}{16}|\vec{\Omega}_0|^4 - \frac{3}{4}|\vec{\Omega}_1|^2 - \vec{\Omega}_0 \cdot \vec{\Omega}_2\right) \frac{\Delta\omega^4}{24} + \dots \\ \hat{p} \cdot \vec{u} &= |\vec{\Omega}_0| \frac{\Delta\omega}{2} + \hat{p} \cdot \vec{\Omega}_1 \frac{\Delta\omega^2}{4} \\ &\quad + \left(\hat{p} \cdot \vec{\Omega}_2 - \frac{1}{4}|\vec{\Omega}_0|^3\right) \frac{\Delta\omega^3}{12} \\ &\quad + \left(\hat{p} \cdot \vec{\Omega}_3 - \frac{3}{2}|\vec{\Omega}_0|\vec{\Omega}_0 \cdot \vec{\Omega}_1\right) \frac{\Delta\omega^4}{48} + \dots \\ \vec{u}_\perp &\equiv \vec{u} - (\hat{p} \cdot \vec{u})\hat{p} \\ &= \vec{\Omega}_{1\perp} \frac{\Delta\omega^2}{4} + \left(\vec{\Omega}_{2\perp} - \frac{1}{2}\vec{\Omega}_0 \times \vec{\Omega}_{1\perp}\right) \frac{\Delta\omega^3}{12} \\ &\quad + \left(\vec{\Omega}_{3\perp} - \vec{\Omega}_0 \times \vec{\Omega}_2 - \frac{1}{2}|\vec{\Omega}_0|^2\vec{\Omega}_{1\perp}\right) \frac{\Delta\omega^4}{48} + \dots \end{aligned} \quad (23)$$

with $\vec{\Omega}_{n\perp} = \vec{\Omega}_n - (\hat{p} \cdot \vec{\Omega}_n)\hat{p}$. The results of [11] are reproduced by substituting $\vec{\tau} = \vec{\Omega}/2$ indicating that the Magnus formula, (20), accurately represents the higher order frequency variation of the Jones (Mueller) matrix.

C. Operator Symmetrization

To recast the Magnus solution of (16) into a product of exponential operators each of which corresponds to a pure n th-order PMD/PDL component, the various noncommuting operators must be correctly symmetrized to preserve the commutation relations between the different matrix orders of $\mathbf{H}_n$ in (16) [3], [19], [20]. The expansions that contain a minimum number of operators up to and including fourth order in $\Delta\omega$ can be derived from the recursive form of the Magnus coefficients, (13), and are given by [22]–[25]

$$\mathbf{M}(\omega)\mathbf{M}(\omega_0)^{-1} = \exp[\mathbf{H}_0\Delta\omega] + O(\Delta\omega^2) \quad (24)$$

$$\begin{aligned} &= \exp\left[\frac{1}{2}\mathbf{H}_0\Delta\omega\right]\exp\left[\frac{1}{2}\mathbf{H}_1\Delta\omega^2\right] \\ &\quad \times \exp\left[\frac{1}{2}\mathbf{H}_0\Delta\omega\right] + O(\Delta\omega^3) \end{aligned} \quad (25)$$

$$\begin{aligned} &= \exp\left[\frac{1}{3}\mathbf{H}_0\Delta\omega\right]\exp\left[\frac{1}{2}\mathbf{H}_1\Delta\omega^2\right] \\ &\quad \times \exp\left[\frac{2}{3}\mathbf{H}_0\Delta\omega\right]\exp\left[\frac{1}{6}\mathbf{H}_2\Delta\omega^3\right] \\ &\quad + O(\Delta\omega^4) \end{aligned} \quad (26)$$

$$\begin{aligned} &= \exp\left[\frac{1}{6}\mathbf{H}_1\Delta\omega^2\right]\exp\left[\frac{1}{4}\mathbf{H}_0\Delta\omega\right] \\ &\quad \times \exp\left[\frac{1}{6}\mathbf{H}_2\Delta\omega^3\right]\exp\left[\frac{1}{4}\mathbf{H}_0\Delta\omega\right] \\ &\quad \times \exp\left[\frac{1}{3}\mathbf{H}_1\Delta\omega^2\right]\exp\left[\frac{1}{2}\mathbf{H}_0\Delta\omega\right] \\ &\quad \times \exp\left[\frac{1}{24}\mathbf{H}_3\Delta\omega^4\right] + O(\Delta\omega^5). \end{aligned} \quad (27)$$

D. Nonsymmetric Model

In Section IV, we compare numerically the accuracy of the symmetrized to the nonsymmetrized Jones matrix operator expansion [2], [26], [27]

$$\mathbf{T}(\omega)\mathbf{T}(\omega_0)^{-1} \approx \mathbf{U}_n \mathbf{U}_{n-1} \dots \mathbf{U}_2 \mathbf{U}_1 \quad (28)$$

in which each matrix

$$\mathbf{U}_n(\Delta\omega) = \exp \left[-\frac{i}{2} \frac{\Delta\omega^n}{n!} \vec{p}_n \cdot \vec{\sigma} \right] \quad (29)$$

yields as in Section II-C an n th-order PMD component with a PMD vector [2]

$$\frac{d\mathbf{U}_n}{d\omega} \mathbf{U}_n^{-1} = -\frac{i}{2} \left[\frac{\Delta\omega^{n-1}}{(n-1)!} \vec{p}_n \right] \cdot \vec{\sigma}. \quad (30)$$

To ensure that $\vec{\Omega}(\omega) \cdot \vec{\sigma} = 2i [d\mathbf{T}/d\omega] \mathbf{T}^{-1}$ to order $\Delta\omega^4$ therefore requires [2]

$$\begin{aligned} \vec{p}_1 &= \vec{\Omega}_0 \\ \vec{p}_2 &= \vec{\Omega}_1 \\ \vec{p}_3 &= \vec{\Omega}_2 + \vec{p}_1 \times \vec{p}_2 \\ \vec{p}_4 &= \vec{\Omega}_3 + \vec{p}_1 \times \vec{p}_3 \\ \vec{p}_5 &= \vec{\Omega}_4 + \vec{p}_1 \times \vec{p}_4 - 3\vec{p}_2 \times (\vec{p}_2 \times \vec{p}_1) + 4\vec{p}_2 \times \vec{p}_3. \end{aligned} \quad (31)$$

III. EMULATOR MODEL

The frequency variation of the polarization vector of an optical system with PMD and PDL can be modeled by dividing the system into a large number of constant birefringent and lossy sections. In each section, the light polarization precesses around the axis of the section's polarization dispersion vector at a rate determined by the magnitude of the birefringence while the attenuation is a maximum and minimum in two generally orthogonal directions.

We consider an optical system with mean PDL $0 \leq \alpha_{\text{avg}} \leq 10$ dB and mean differential group delay (DGD) $\tau_{\text{avg}} = 20$ ps as calculated with 0 dB mean PDL. To represent this by an $N_{\text{sec}} = 100$ section emulator, we set the frequency-independent PMD vector magnitude of the m th section and the differential attenuation between the most and least attenuated polarization states to $|\vec{\tau}_m| = \tau_{\text{avg}} \sqrt{3\pi/8N_{\text{sec}}} = 2.17$ ps and $0 \leq |\vec{\alpha}_m| \leq 1$ dB, respectively. The orientation of the birefringent and attenuation axes of the m th section, specified by the unit vectors $\vec{\tau}_m/|\vec{\tau}_m|$ and $\vec{\alpha}_m/|\vec{\alpha}_m|$, respectively, are selected from a uniform statistical distribution over the Poincaré sphere.

Denoting the Jones and complex 3×3 Stokes matrices of the m th emulator section, $m = 1, 2, \dots, N_{\text{sec}}$, by

$$\begin{aligned} \mathbf{V}_m(\omega) &= \exp \left[\frac{1}{2} \vec{\alpha}_m \cdot \vec{\sigma} \right] \exp \left[-\frac{i}{2} \omega \vec{\tau}_m \cdot \vec{\sigma} \right] \\ \mathbf{R}_m(\omega) &= \exp [i\vec{\alpha}_m \times] \exp [\omega \vec{\tau}_m \times] \end{aligned} \quad (32)$$

the corresponding complex principal state vector, $\vec{V}_m \cdot \vec{\sigma} = 2i [d\mathbf{V}_m/d\omega] \mathbf{V}_m^{-1}$, of the m th section can be written as

$$\vec{V}_m = \exp [i\vec{\alpha}_m \times] \vec{\tau}_m. \quad (33)$$

Here, for any real valued ψ and unit vector $\hat{r}$, $\exp [i\psi \hat{r} \times] = \cosh \psi \mathbf{I}_3 + (1 - \cosh \psi) \hat{r} \hat{r}^T + i \sinh \psi (\hat{r} \times)$ [1].

By induction, the n th derivative, $n \geq 1$, of the total complex principal state vector after m emulator sections, $\vec{W}_m = \vec{V}_m + \mathbf{R}_m \vec{W}_{m-1}$, is given by

$$\begin{aligned} \frac{d^n \vec{W}_m}{d\omega^n} &= \mathbf{R}_m \frac{d^n \vec{W}_{m-1}}{d\omega^n} \\ &\quad - \sum_{k=1}^n \binom{n}{k} [-\vec{V}_m \times]^k \frac{d^{n-k} \vec{W}_m}{d\omega^{n-k}} \end{aligned} \quad (34)$$

with $\vec{W}_0 = 0$. Analogously, setting $\mathbf{T}_0 = \sigma_0$

$$\begin{aligned} \frac{d^n \mathbf{T}_m}{d\omega^n} &= \mathbf{V}_m \frac{d^n \mathbf{T}_{m-1}}{d\omega^n} \\ &\quad - \sum_{k=1}^n \binom{n}{k} \left[\frac{i}{2} \vec{V}_m \cdot \vec{\sigma} \right]^k \frac{d^{n-k} \mathbf{T}_m}{d\omega^{n-k}} \end{aligned} \quad (35)$$

yields the n th derivative of the Jones matrix $\mathbf{T}_m = \mathbf{V}_m \mathbf{T}_{m-1}$ [28]. Both (34) and (35) can be easily implemented numerically. Further, in accordance with the Taylor expansion of the Jones matrix in orders of $\vec{\Omega}(\omega)$ presented in Section II-B, (35) will be subsequently employed in the numerical evaluation of (23).

IV. RESULTS

A. PMD Model Comparison

We now compare the procedures of Section II for zero PDL through an analysis of the maximum error, $\epsilon_\kappa(\Delta\omega)$, between the estimated $\mathbf{T}_{\text{est}}(\omega)$ and the emulated Jones matrices $\mathbf{T}(\omega)$ over the frequency interval $|\omega - \omega_0| \leq \Delta\omega$. This error is given for each realization $\kappa = 1, 2, \dots, (N_\kappa = 10^6)$ of the emulator by

$$\epsilon_\kappa(\Delta\omega) = \max_{|\omega - \omega_0| \leq \Delta\omega} \left[\frac{\|\mathbf{T}_{\text{est}}(\omega) - \mathbf{T}(\omega)\|}{\|\mathbf{T}(\omega)\|} \right] \cdot 100\% \quad (36)$$

where $\|\mathbf{T}\| \equiv \sqrt{\text{Trace}(\mathbf{T}^\dagger \mathbf{T})}$ is the Euclidean matrix norm. For each emulator realization κ the bandwidth $\Delta\omega_\kappa$ is found with a bisection method [29] applied to the condition that $\epsilon_\kappa(\Delta\omega_\kappa) = \epsilon_{\text{max}}$ for a maximum specified error ϵ_{max} . The probability density function (pdf) of $\Delta\omega_\kappa$ is then estimated for each of the numerical techniques of Section II from the N_κ emulator realizations. Note that (36), unlike the Q value or bit-error rate, is independent of a specific system implementation and, therefore, provides an unambiguous measure of the algorithmic precision [28].

The pdf of $\Delta\omega_\kappa$ for $\epsilon_{\text{max}} = 2\%$ as a function of the normalized frequency $\Delta\nu = \tau_{\text{avg}} \Delta\omega_\kappa / 2\pi$ is displayed in Fig. 1, in which $\mathbf{T}_{\text{est}}(\omega)$ is approximated by the fourth-order versions of the Magnus expansion (20) (o), (23) (●), the symmetric expansion of (27) (×) and the nonsymmetric model, (28) (+).

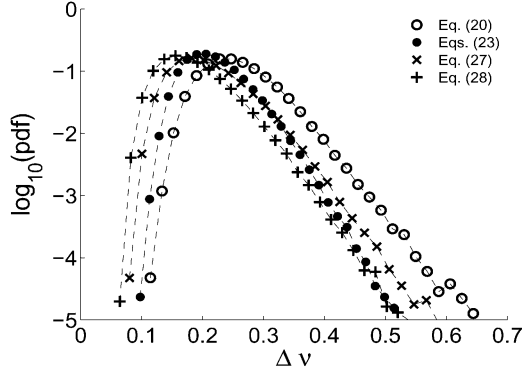


Fig. 1. The pdf of the normalized bandwidth, $\Delta\nu = \tau_{\text{avg}}\Delta\omega_\kappa/2\pi$, for $\epsilon_\kappa(\Delta\omega_\kappa) = \epsilon_{\text{max}} = 2\%$ evaluated from $N_\kappa = 10^6$ realizations of a $N_{\text{sec}} = 100$ section fiber emulator with 0 dB PDL as determined by the fourth order Magnus expansion (20) (○), (23) (●), the symmetric product of (27) (×) and the nonsymmetric expansion of (28) (+).

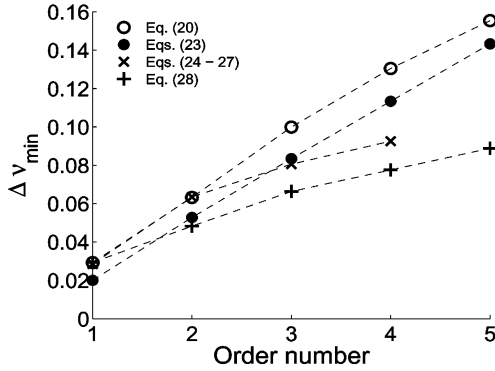


Fig. 2. Smallest normalized bandwidth, $\Delta\nu_{\text{min}}$, for which the maximum observed error of (36) is $< 2\%$ for 99.99% of the randomly generated emulator realizations versus the expansion order, N , for the Magnus expansion of (20) (○), the symmetric operator expansion of (24)–(27) (×), the nonsymmetric expansion of (28) (+), and (23) (●).

Here we have displayed every third data point to improve legibility. Clearly, the frequency range over which the Jones matrix can be accurately estimated is greatest for the Magnus expansion. However, (27) yields more precise results than (28) though within a more restricted frequency range than that of (20).

To determine the variation of the frequency range over which each model accurately represents the Jones matrix with the expansion order N , we determine as in [28] the normalized frequency $\Delta\nu_{\text{min}}$ such that the probability $P(\Delta\nu \geq \Delta\nu_{\text{min}})$ of $\Delta\nu$ exceeding $\Delta\nu_{\text{min}}$ is 99.99% for values of N between 1 and 5. Fig. 2 then displays $\Delta\nu_{\text{min}}$ as a function of N in which the ○, ×, + and ● markers indicate the Magnus expansion of (20), the symmetric model of (24)–(27), the nonsymmetric model (28) and (23), respectively. As a result of statistical fluctuations, the modeling error exceeds ϵ_{max} with a probability $1 - P(\Delta\nu \geq \Delta\nu_{\text{min}})$. However, we have previously established that the variation of $\Delta\nu_{\text{min}}$ with expansion order N is qualitatively independent of $P(\Delta\nu \geq \Delta\nu_{\text{min}})$ and ϵ_{max} [28]. Therefore, while a $\Delta\nu_{\text{min}}$ cannot be specified such that $\Delta\nu \geq \Delta\nu_{\text{min}}$ for all possible emulator realizations, for values of $\Delta\nu$ less than ≈ 0.1 the pdf decreases rapidly in Fig. 1 indicating that slightly lowering $\Delta\nu_{\text{min}}$ substantially reduces the probability that the estimation error exceeds ϵ_{max} .

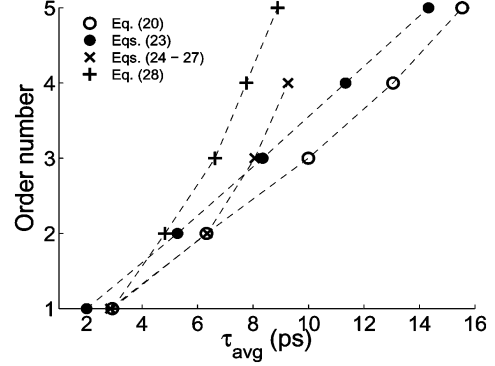


Fig. 3. Same as Fig. 2, but for the number of expansion orders required to determine the frequency evolution of the Jones matrix with a maximum error of $\epsilon_{\text{max}} = 2\%$ with 99.99% confidence over a $|\omega - \omega_0|/2\pi \leq 10$ GHz frequency interval for mean DGD τ_{avg} and 0 dB mean PDL.

In Fig. 3, we display as a function of τ_{avg} the number of expansion orders required for each of the models of Section II to reproduce the frequency evolution of the Jones matrix to within $\epsilon_{\text{max}} = 2\%$ for $P(\Delta\nu \geq \Delta\nu_{\text{min}}) = 99.99\%$ over a $|\omega - \omega_0|/2\pi \leq 10$ GHz frequency interval. While the range of τ_{avg} for the desired level of accuracy is greatest for the Magnus expansion, (20), even for relatively low values of τ_{avg} many terms in the Taylor expansion of the PMD vector are required to insure sufficient precision over the entire 10 GHz frequency interval. Reducing the confidence level $P(\Delta\nu \geq \Delta\nu_{\text{min}})$ or increasing the acceptable modeling error ϵ_{max} , however, decreases the expansion order required to attain a given value of τ_{avg} .

B. Magnus Expansion With PMD and PDL

Next, in Fig. 4, we examine the relative accuracy of the Magnus expansion in the presence of PDL by displaying the variation of $\Delta\nu_{\text{min}}$ with the mean PDL, α_{avg} , for a fiber emulator model after truncating (20) to first (○), second (×), third (+), fourth (●), and fifth (△) order in $\Delta\omega$. Here we have employed a constant segment DGD of $|\vec{\tau}_m| = 2.17$ ps for $m = 1, \dots, N_{\text{sec}}$. The normalized bandwidth $\Delta\nu = \tau_{\text{avg}}\Delta\omega_\kappa/2\pi$ in Fig. 4 accounts for the slight dependence of τ_{avg} on α_{avg} , arising from the relationship $\tau_{\text{avg}} = \langle \text{Re} \{ \sqrt{\vec{W} \cdot \vec{W}} \} \rangle$ for nonzero PDL, where $\langle \dots \rangle$ represents an ensemble average over emulator realizations [5]. Although the bandwidth reduction at large α_{avg} is most evident at high expansion orders, for $\alpha_{\text{avg}} \leq 4$ dB the worst case reduction is $\approx 7 \times 10^{-3}$ in $\Delta\nu_{\text{min}}$ up to fifth order, implying that a nominal 7.6 GHz bandwidth would be reduced by 0.35 GHz for $\tau_{\text{avg}} = 20$ ps and by ≈ 0.042 or 2.1 GHz for $\alpha_{\text{avg}} = 10$ dB.

Finally, in Fig. 5, we display as in Fig. 3 the variation of the number of Magnus expansion orders required to reproduce the frequency evolution of the Jones matrix to within $\epsilon_{\text{max}} = 2\%$ for $P(\Delta\nu \geq \Delta\nu_{\text{min}}) = 99.99\%$ over a $|\omega - \omega_0|/2\pi \leq 10$ GHz frequency interval with τ_{avg} , for values of mean PDL in the range $0 \leq \alpha_{\text{dB}} \leq 10$ dB. Evidently, relative to $\alpha_{\text{avg}} = 0$ dB, the mean DGD must be reduced by approximately 4.2 ps to limit sufficiently the modeling error for $\Delta\omega/(2\pi) = 10$ GHz for the worst case of 9.9 dB mean PDL and $N = 5$ expansion orders. In

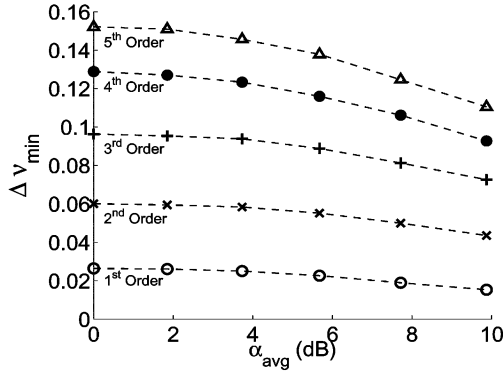


Fig. 4. Variation of $\Delta\nu_{\min}$ with the mean PDL, α_{avg} , of a $N_{\text{sec}} = 100$ section emulator for the Magnus expansion of (20) truncated to first (o), second (+), fourth (•), and fifth (Δ) order in $\Delta\nu$.

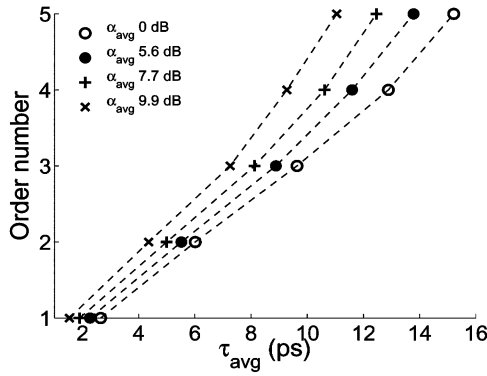


Fig. 5. Same as Fig. 3, but for the number of Magnus expansion orders required to limit the error in the computed frequency evolution of the Jones matrix to $\epsilon_{\text{max}} = 2\%$ with 99.99% confidence over a $|\omega - \omega_0|/2\pi \leq 10$ GHz frequency interval for a mean DGD τ_{avg} and values of the mean PDL in the range $0 \leq \alpha_{\text{avg}} \leq 10$ dB.

an analogous fashion, for $\alpha_{\text{avg}} \leq 4$ dB τ_{avg} should be reduced by ≈ 0.7 ps.

C. Taylor Expansion Estimation Bandwidth

We next examine the width of the frequency region over which the truncated Taylor expansion

$$\vec{W}_{\text{est}}(\omega) = \sum_{n=0}^N \vec{W}_n \frac{\Delta\omega^n}{n!} \quad (37)$$

accurately reproduces the frequency evolution of $\vec{W}(\omega)$. Here we first construct the 3×3 statistical correlation matrix $\rho_{ab} = \langle W_a W_{\text{est},b}^* \rangle / (\sigma_a \sigma_{\text{est},b})$, $a, b = x, y, z$, formed from the components $W_a(\omega)$ and $W_{\text{est},b}(\omega)$ of the $\vec{W}(\omega)$ and $\vec{W}_{\text{est}}(\omega)$ vectors where $\langle \dots \rangle$ represents an ensemble average over 10^5 emulator realizations. Since $\langle W_a \rangle = \langle W_{\text{est},b} \rangle = 0$ and W_a is statistically uncorrelated with $W_{\text{est},b}$ for $a \neq b$, $\sigma_a^2 = \langle |W_a|^2 \rangle$ and $\sigma_{\text{est},b}^2 = \langle |W_{\text{est},b}|^2 \rangle$ while ρ_{ab} is proportional to the identity matrix. Accordingly, we consider only the frequency variation of $\rho_{xx} = \langle W_x W_{\text{est},x}^* \rangle / (\sigma_x \sigma_{\text{est},x})$. In Fig. 6, we plot the correlation $|\rho_{xx}|$ as a function of normalized frequency $\Delta\nu = \tau_{\text{avg}} \Delta\omega / 2\pi$ for expansion orders $N = 0, 2, 4, \dots, 20$ where $\vec{W}_n$

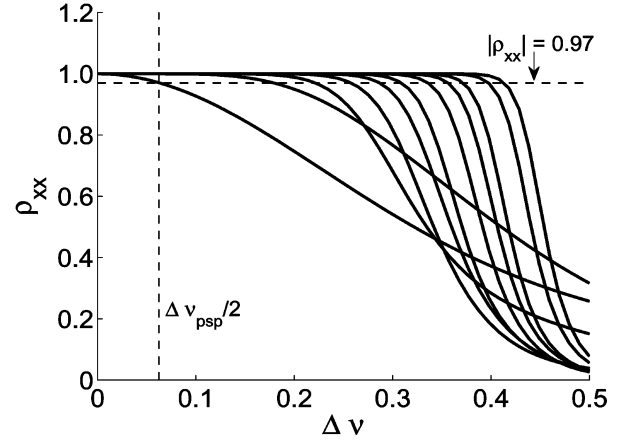


Fig. 6. Variation with normalized frequency, $\Delta\nu = \tau_{\text{avg}} \Delta\omega / 2\pi$, of the correlation $|\rho_{xx}|$ between the x -components of $\vec{W}$ and $\vec{W}_{\text{est}}$ for Taylor orders $N = 0, 2, 4, \dots, 20$ and 0 dB mean PDL. A vertical dotted line indicates the half bandwidth of the principal state, $\Delta\nu_{\text{psp}}/2$, with $\Delta\nu_{\text{psp}} = 1/8$.

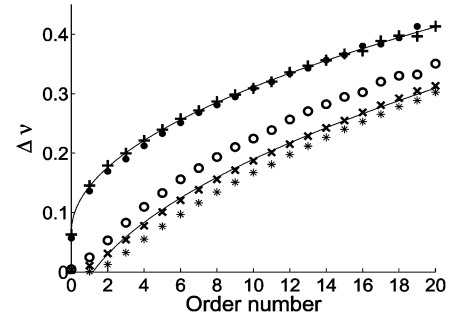


Fig. 7. Variation of the frequency interval $\Delta\nu$ for which $|\rho_{xx}| \geq 0.97$ (+ markers) and $|\rho_{xx}| \geq 0.9999$ (o markers) with the Taylor expansion order, N and 0 dB mean PDL. Analogous results for $|\rho_{xx}| \geq 0.97$ and 9.9 dB mean PDL are indicated by the • markers. Also shown is $\Delta\nu_{\min}$, as calculated with the numerical methods of Section IV-A for $\epsilon_{\text{max}} = 2\%$ and $P(\Delta\nu \geq \Delta\nu_{\min}) = 99.99\%$, with 0 dB (x markers) and 9.9 dB (* markers) mean PDL. The solid lines are optimal fits to square-root functions.

is calculated from (34) with 0 dB mean PDL. We then employ $|\rho_{xx}| = 0.97$, which results from a $N = 0$ Taylor expansion calculation performed according to Section IV-A at the half-bandwidth, $\Delta\nu_{\text{psp}}/2 = 1/16$, of the principal state. This yields the + markers in Fig. 7 for the frequency interval $\Delta\nu$ for which $|\rho_{xx}| \geq 0.97$ as a function of the expansion order. Including PDL only moderately reduces the bandwidth for which $|\rho_{xx}| \geq 0.97$ as indicated by the • markers in Fig. 7 which employ $\alpha_{\text{avg}} = 9.9$ dB.

In a second calculation, shown as x markers in Fig. 7, we apply the numerical techniques of Section IV-A to identify the frequency interval $\Delta\nu_{\min}$ for 0 dB mean PDL over which the maximum error

$$\epsilon(\Delta\omega) = \max_{|\omega - \omega_0| \leq \Delta\omega} \left[\frac{\|\vec{W}_{\text{est}}(\omega) - \vec{W}(\omega)\|}{\|\vec{W}(\omega)\|} \right] \cdot 100\% \quad (38)$$

equals 2%, while $P(\Delta\nu \geq \Delta\nu_{\min}) = 99.99\%$ (the solid lines of Fig. 7 are least-square fits of the results of both calculations to square-root functions [30]). The analogous results for $\alpha_{\text{avg}} = 9.9$ dB, marked * in Fig. 7, depend only weakly on the mean PDL of the emulator. Evidently, limiting the maximum

TABLE I
MEASURED RELATIVE ERROR $\epsilon_{\text{exp}}(\%)$

DGD PDL	2.2 ps 0.3 dB	2.3 ps 1.0 dB	3.9 ps 2.0 dB	9.0 ps 0.3 dB	6.8 ps 0.9 dB	10.0 ps 2.3 dB
$\mathbf{N}_1$	0.21	0.33	0.54	0.41	0.48	0.69
$\mathbf{N}_2$	4.3	4.5	6.3	4.8	6.0	7.3
$\mathbf{N}_3$	6.8	9.7	17.4	19.0	14.4	—

modeling error reduces the bandwidth compared to requiring a minimum statistical correlation between the estimated and exact complex principal state vectors. This is clearly evident from the $|\rho_{xx}| = 0.9999$ results for $\alpha_{\text{avg}} = 0$ dB in Fig. 7 (○ markers), which more accurately reproduce the $\epsilon_{\text{max}} = 2\%$ curve.

D. Experimental Verification

Finally, we have verified the experimental relevance of the Magnus expansion for the first three orders $n = 1, 2, 3$ in $\Delta\omega$ in the presence of both PMD and PDL. Measurements were taken with a HP 81689A tunable laser source continuously calibrated with an Ando AQ6140 wavelength meter in 0.1 nm steps over a 4 nm range while the output Jones matrix $\mathbf{T}(\omega)$ was measured by a HP 8509B polarimeter. The device under test (DUT) consisted of three sections of PM fiber interleaved with two adjustable PDL elements, yielding PDL values from 0 to 2 dB as measured at the HP 8509B.

After a Jones to Mueller matrix conversion according to (2), the matrix exponent at each frequency was extracted according to $\mathbf{N}(\omega_m - \omega_0) = \ln[\mathbf{M}(\omega_m)\mathbf{M}(\omega_0)^{-1}]$, in which $m = 1, 2, \dots, N_\omega$ and N_ω represents the number of measured frequencies. The Taylor series coefficients, denoted by $\mathbf{N}_n^{(1)}$ were then obtained from a polynomial least square fit to $\mathbf{N}(\Delta\omega)$. For comparison, the analytic expression, (16), $\mathbf{N}_n^{(2)}$ was also evaluated after constructing $\tilde{\Omega}_n$ and $\tilde{\Lambda}_n$ from a finite-difference approximation to the measured Jones matrices, $\mathbf{T}(\omega_m)$, according to $\tilde{W} \cdot \tilde{\sigma} = 2i[d\mathbf{T}/d\omega]\mathbf{T}^{-1}$. The corresponding measured relative errors for system configurations with different measured DGD and PDL values at the center frequency from $\epsilon_{\text{exp}} = 100\% \times \left\| \mathbf{N}_n^{(2)} - \mathbf{N}_n^{(1)} \right\| / \left\| \mathbf{N}_n^{(1)} \right\|$ are summarized in Table I. While wavelength instability and other systematic errors distorted our estimate of $\mathbf{N}_3$ for the 0.1-nm wavelength step size, $\mathbf{N}_1$ and $\mathbf{N}_2$ are accurate to within 8% for PDL values ≤ 2 dB. Accordingly, despite the limited number of system realizations, the measured frequency dependence of the Mueller matrix is consistent with the predictions of (16) to at least second order.

V. CONCLUSION

While our Magnus expansion model of the frequency dependence of the Mueller matrix in linear optical systems affected by both PMD and PDL reproduces previous power-series expansion and differential equation solution techniques, it additionally preserves the symmetry of the Mueller matrix in all expansion orders. This yields physically realizable operator expansions that facilitate the design of joint PMD and PDL compensators [3], [19], [20]. Other practically relevant models of PMD frequency variation [19], [26] can be obtained by applying the Baker–Campbell–Hausdorff (BCH) identity [3], [21] to our

novel low order expansions of (16) while improved accuracy within a broader frequency range could potentially be achieved by extending the formalism according to [9], [31]. Unfortunately, while, in principle, numerous terms of the Magnus expansion can be calculated to increase numerical precision, currently our ability to measure third and higher order PMD vectors experimentally is limited.

Although the frequency behavior of the Jones or Mueller matrix is generally characterized through Taylor expansion coefficients, a fifth order Magnus expansion is, for example, required to represent the Jones matrix accurately for a mean DGD of 15 ps. Thus, other parameterizations may be preferable for large mean DGD fibers [2], [20], [32]. Numerical models such as the Jones matrix expansions and the interpolation procedures of [28] that describe a large span of frequencies can as well lead to enhanced numerical accuracy and programming simplicity for high bit-rate single or multiple channel optical system simulations. Finally, although PMD model performance is quantified in this paper by the maximum error incurred within a given frequency range, the system penalty due to PMD and PDL in an optical network may be a highly nonlinear function of this error, dependent upon implementation details [33], [34], or may possibly be sensitive only to the coefficients of the low Magnus expansion orders. A full analysis of system performance with the Magnus expansion, however, clearly exceeds the scope of the present paper.

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David Yevick, (SM'86–F'96) photograph and biography not available at the time of publication.

Derek Dumas, photograph and biography not available at the time of publication.